

Proteomics Explorer: User Manual & Technical Guide

Version: 1.2.0 **Author:** Simon Weaver (Notre Dame Mass Spectrometry and Proteomics Facility) **License:** MIT

Table of Contents

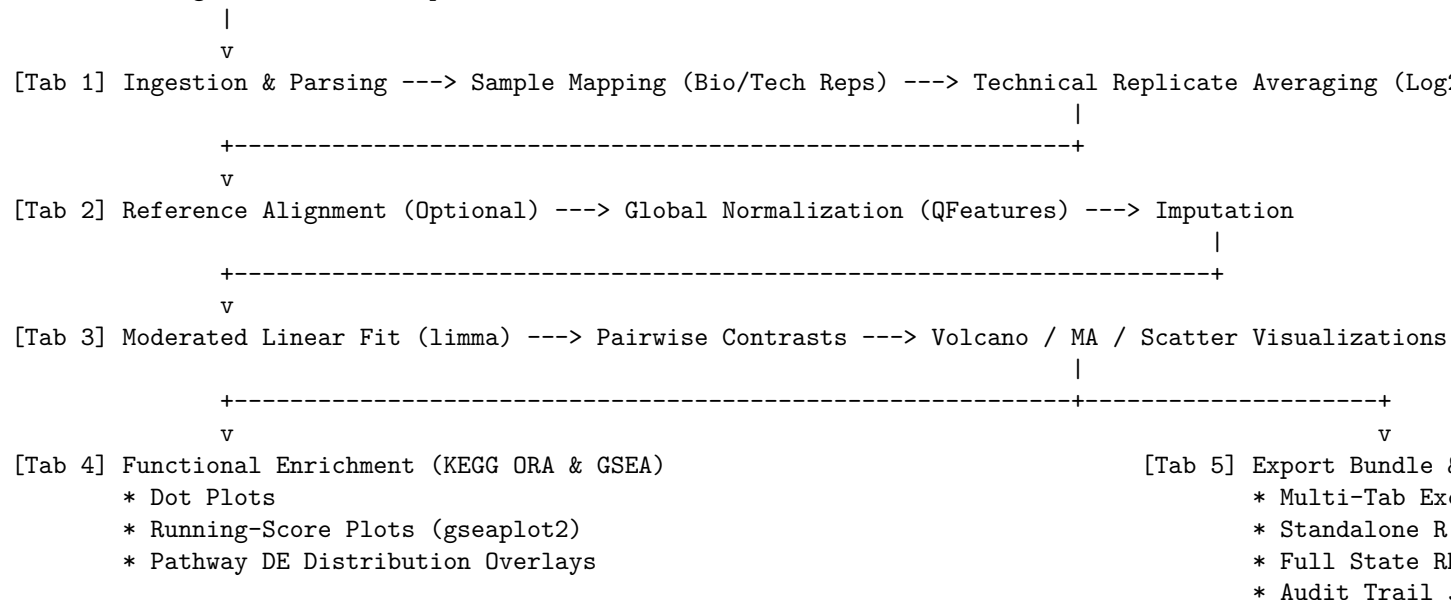
1. Overview & Architecture
 2. Prerequisites & Launch Options
 3. Tab 1: Data Import & Metadata Setup
 4. Tab 2: Normalization & Imputation
 5. Tab 3: Differential Expression & Visualization
 6. Tab 4: Functional Pathway Enrichment (KEGG & GSEA)
 7. Tab 5: Export & Reproducibility Audit
 8. Troubleshooting & FAQs
-

1. Overview & Architecture

Proteomics Explorer is an interactive R/Shiny platform built for quantitative bottom-up and label-free mass spectrometry (LFQ/DIA) analysis. It integrates data ingestion across major search platforms, customizable normalization pipelines, missing value imputation, limma-powered moderated linear modeling, and functional pathway enrichment via clusterProfiler.

Pipeline Flow

Raw Search Engine / Matrix Output



2. Prerequisites & Launch Options

Prerequisites

- R ($\geq 4.2.0$) installed.
- Active internet connection on initial launch to retrieve CRAN and Bioconductor packages.
- Dependencies are sandboxed into a local `app_lib/` folder to prevent library permission conflicts.

Option A: 1-Click Launch (Windows)

Double-click `launch_app.bat` in the repository root directory. The batch script will automatically: 1. Detect your local 64-bit R installation from PATH, Program Files, or Windows Registry. 2. Set the user library variable to `./app_lib`. 3. Run `launch.R` to check and install missing packages, then open the Shiny app in your default browser.

Option B: Terminal / RStudio (macOS, Linux, Windows)

From an active R session in the project directory, run:

```
source("launch.R")
```

3. Tab 1: Data Import & Metadata Setup

3.1 Supported File Formats

Select the appropriate format from the Select Source Format dropdown:

Engine / Platform	Expected Input File	Key Identifier Columns	Quantitative Parsing Logic
Spectronaut	<code>.tsv</code> / <code>.txt</code> Export Report	PG.ProteinAccessions, PG.Genes, PG.ProteinDescriptions	Extracts PG.Quantity per R.Condition + R.Replicate; converts values ≤ 0 to NA and applies $\log_2$.
MSFragger	<code>combined_protein.tsv</code>	Protein.ID (or Protein), Gene, Description	Reads *.Intensity columns (prioritizes standard intensity, falls back to MaxLFQ Intensity if standard is missing).
DIA-NN	<code>report.pg_matrix.tsv</code>	Protein.Group, Genes, First.Protein.Description	Parses sample raw data paths, automatically strips long directory paths and file extensions (<code>.raw</code> , <code>.wiff</code> , <code>.d</code>).

Engine / Platform	Expected Input File	Key Identifier Columns	Quantitative Parsing Logic
PEAKS	db.proteins.csv or lfq.proteins.csv	Accession, Description	Extracts gene symbols using regex (GN=([^\s]+)); strips trailing pipe notations (
Generic Table	.tsv / .csv	Accession, Protein.ID, or Protein	Auto-detects Long vs. Wide format. In Wide format, quant columns must have .Intensity or numeric values.
Saved State	*_state.rds	Serialized RDS list	Bypasses raw ingestion and directly restores entire analysis state.

3.2 Sample Metadata Mapping

Once the file is uploaded, the app auto-populates sample mapping inputs: * Condition: Biological experimental group (e.g., WT, KO, Treated). * Bio Rep: Biological replicate number (1, 2, 3).

3.3 Technical Replicate (TR) Averaging

If multiple injections were performed on the same biological samples: 1. Check the box “Dataset contains Technical Replicates (Injections)”. 2. A third input column (Tech Rep) will appear. 3. For each unique biological unit (Condition + BioRep), technical replicates are averaged on the log2 scale: $\text{LogInt_Consolidated} = \text{mean}(\text{LogInt_valid})$ * Missing Values (NAs): If a protein is detected in 1 out of 2 technical injections, that single valid value is retained. If all technical injections are missing, the biological sample is marked as NA. 4. Click “Process & Map Samples” to register the design.

4. Tab 2: Normalization & Imputation

4.1 Step 1: Reference Protein Normalization (Optional)

Used for spike-ins, biological internal standards, or cell lysis controls (e.g., GAPDH, ACTB, or a specific accession): 1. Check “Normalize to Reference Protein / Loading Control”. 2. Select “Gene Symbol” or “Protein Accession”, and type the query. 3. For each sample s, the intensity offset is calculated as: $\text{Offset_s} = \text{Ref_s} - \text{mean}(\text{Ref_global})$ $\text{Intensity_corrected_s} = \text{Intensity_s} - \text{Offset_s}$

4.2 Step 2: Global Normalization (QFeatures)

Aligns whole-proteome sample distributions: * None: Retains raw log2 measurements. * Median Centering (center.median): Subtracts each sample median intensity to align medians at zero. * Mean Centering (center.mean): Subtracts each sample mean intensity to align means at zero. * Quantile Normalization (quantiles): Enforces identical distribution curves across all runs. * Variance Stabilizing (vsn): Calibrates affine transformations to stabilize variance across the dynamic range.

4.3 Step 3: Missing Value Imputation

- Hybrid (MAR: KNN / MNAR: MinDet):
 - Missing at Random (MAR): Proteins present in $\geq 50\%$ of replicates within a condition are imputed using global neighbor averaging (nbavg/KNN).
 - Missing Not at Random (MNAR): Proteins present in $< 50\%$ of replicates represent dropouts below detection limits and are imputed condition-wise via deterministic minimum substitution (MinDet, positioned at $\text{Min_observed} - 0.5$).
- No Imputation (Show 1-Condition Dropouts on Margins): No synthetic values are generated. Proteins completely missing in one condition but observed in another are retained and projected along plot margins at fixed offsets.
- Standard Methods: Full matrix imputation using MinDet, knn, nbavg, or None.

Click “Apply Transformation” to execute the pipeline and update the interactive QC density plots.

5. Tab 3: Differential Expression & Visualization

5.1 Model Fitting & Contrast Evaluation

Differential expression is computed automatically using limma linear modeling: * The model matrix is fit without an intercept ($\sim 0 + \text{Condition}$). * Pairwise contrasts (e.g., $\text{CondB} - \text{CondB}$) are built and tested using Empirical Bayes moderation (eBayes). * Invert Contrast Direction: Instantly flips $\log_2$ fold-changes ($\log_{FC} \rightarrow -\log_{FC}$) and swaps experimental/reference sample assignments across plots and tables without refitting the linear model.

5.2 Plot Layouts

- Volcano Plot: Plots $\log_2(\text{Fold Change})$ vs. $-\log_{10}(\text{Adjusted p-value})$.
- MA Plot: Plots average intensity (AveExpr) vs. $\log_2(\text{Fold Change})$. 1-condition dropouts are pinned along visual dashed margins ($y = \pm y_{\text{cap}}$).
- Scatter Plot: Plots $\log_2$ condition means directly (Ref vs. Exp), with dropouts positioned along the lower baseline.

5.3 Collapsible Accordion Views

- Interactive Plotly Explorer (Top): Pan, zoom, and mouse-over points to inspect gene symbols, accessions, and statistics.
- Publication-Ready ggplot (Bottom): Rendered with ggrepel text labeling and publication-quality styling.
- Export individual plots as publication-ready PNG (300 DPI) or vector PDF files with customizable dimensions.

5.4 Custom Protein Highlighting

Highlight biological pathways, complexes, or candidate genes: 1. Click “+ Add Protein Set”. 2. Choose “List” (comma-, space-, or newline-separated entries) or “Regex Match” (e.g., ^HIST. , ^RPL.). 3. Assign custom colors and toggle highlighting and labels independently.

6. Tab 4: Functional Pathway Enrichment (KEGG & GSEA)

Analyze functional enrichment using clusterProfiler for Mouse (*Mus musculus*) or Human (*Homo sapiens*).

6.1 Dual Ingestion Modes

- Use Active App Analysis: Runs enrichment on the contrast currently selected in Tab 3 (respecting contrast direction inversion).
- Upload Excel Export (.xlsx): Ingests previously generated Excel workbooks to run enrichment without reloading raw MS runs.

6.2 Analysis Methods

- KEGG Over-Representation Analysis (ORA): Evaluates significant proteins against the measured proteome background using Fisher exact test. Filter by direction: All Significant, Upregulated Only, or Downregulated Only.
- Ranked GSEA: Uses a ranked list of all measured proteins sorted by Log2(Fold Change). Generates split-sign dot plots showing activated vs. suppressed pathways.

6.3 Pathway Distribution Inspection

- Running Score Curve: View gseaplot2 profiles for any selected pathway.
 - Pathway DE Distribution Plot: Overlays all proteins from a specific pathway directly onto the Volcano, MA, or Scatter plot, allowing you to see the individual protein fold changes driving that pathway score.
-

7. Tab 5: Export & Reproducibility Audit

7.1 Export Deliverables

- Excel Results Workbook (.xlsx): Multi-sheet workbook with one tab per comparison. Column names dynamically incorporate condition labels (e.g., Log2_Treated_Mean, Log2_Control_Mean, Treated_Valid_Count, Control_Valid_Count).
 - Reproducible R Pipeline Script (.R): Generates a self-contained R script recreating all file parsing, metadata mappings, normalization transformations, limma designs, and publication figures without needing the Shiny UI.
 - Full Session State (.rds): A serialized snapshot containing all raw data, sample maps, normalized matrices, and DE tables. Can be reloaded into Tab 1 under “Re-import Exported RDS/State” to resume analysis instantly.
 - JSON Audit Trail (.json): Tracks the application version, session timestamp, R version, loaded package versions, user mapping dictionary, normalization settings, and imputation counts.
-

8. Troubleshooting & FAQs

- Error: “No valid intensity columns found”
 - MSFrager: Ensure your table has columns ending with Intensity or MaxLFQ Intensity.

- Spectronaut: Ensure PG.Quantity is present in your exported report.
- Generic Table: Select Generic Table (Long or Wide) and verify your intensity columns end with .Intensity or are numeric.
- Why are my fold changes backwards?
 - Contrasts are built as Condition1 - Condition2 based on alphanumeric ordering. To swap the direction without refitting your model, check “Invert Contrast Direction” in Tab 3.
- How do I prevent Git from tracking large package downloads?
 - Verify that `app_lib/` is listed in your `.gitignore`. If it was previously staged, run `git rm -r --cached app_lib` in your terminal.
- Can I run this without the GUI?
 - Yes. Download the standalone script via “Download Reproducible R Script” in Tab 5. It runs headless from any standard R environment.